KNN deadline 30/03

- **optional**
- **not recommended to work in groups, instead if you do it in distance**
- **updated TP version on Moodle**

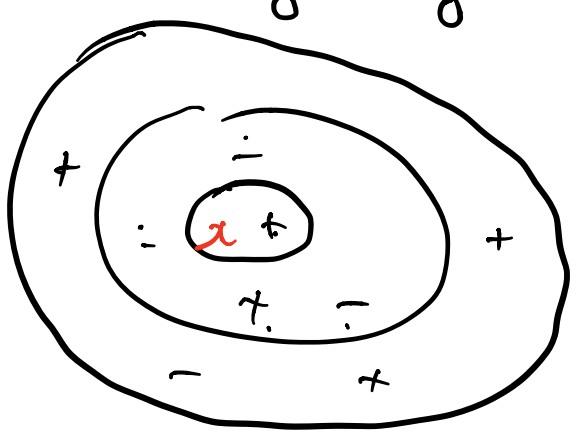
TRN

stock data $[\{x_i, y_i\}]$

TST

$\forall$ instance in TST

- calculate distance with all TRN instances
- pick k neirest
- return the most frequent class among neighbors



$k = 1 \quad x \rightarrow +$

$k = 5 \quad \left.\begin{matrix} 2\ + \\ 3\ - \end{matrix}\right\} \Rightarrow x \rightarrow -$

Ex 2. $k = [1, 3, 5, 10, 20, 50]$

Euclidean $D = [(x_i - x_j)^T (x_i - x_j)]^{1/2}$

$= \left(\sum_{k=1}^{d} (x_{ik} - x_{jk})^2\right)^{1/2}$

Manchatan $D = \sum_{k=1}^{d} |x_{ik} - x_{jk}|$

Chabyshev $D = \max_k |x_{ik} - x_{jk}|$

Machalanobis $D = [(x_i - x_j)^T \Sigma^{-1} (x_i - x_j)]^{1/2}$

$= \sum_{kl}^{d} (x_{ik} - x_{jk}) \underline{\Sigma^{-1}_{kl}} (x_{il} - x_{jl})]^{1/2}$

$\rightarrow \Sigma = I\sigma$, $\sigma = \frac{1}{d} \sum_{k=1}^{d} \sigma_k$

$$\rightarrow \Sigma = \begin{bmatrix} \sigma_1 & & & 0 \\ & \sigma_2 & & \\ & & \ddots & \\ 0 & & & \sigma_k \end{bmatrix}$$

$\rightarrow \Sigma$ full cov matrix

Ex1 CIRAF-10

Euclidean distance, $K=1$

→ 2 for loops

→ 1 for loop

→ no for loop

(expect approx accuracy ≈ 27% for both $K=1$, $K=5$)

Ex3 decision surfaces

(comment / Explane how the performance changes wrt diff values of k.)

Ex 4 Linear Transformation

$\begin{bmatrix} x \\ y \end{bmatrix} [x \ y] = \begin{bmatrix} 1 & \\ & 1 \end{bmatrix}$ ← $x^T x = 1$ → unit circle

$x^2 + y^2 = 1$

$x^T A x$ transform it → no a circle anymore

$z^T A z = 1$ solve eq. and draw new cicle

↳ Problem when import imread

Go data_utils.py

comment from scipy.misc ...

uncomment from imageio ...

↳ Problem with CIFAR-10

If you cannot import it

do ex1 using iris data set